ECE 3340 Numerical Methods

Homework 2: Prerequisite Programming

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Solve the following problems from **Chapter 2, Prerequisites: Programming**. Use any available space to work out the problem and **place your final solution in the box provided**.

Problem 1: Provide the contents of the registers corresponding to w, x, y, and z

```
...
unsigned char a = 200;
unsigned char b = 2;
unsigned int c = 128;
unsigned int w = c % 7
unsigned int x = a + c;
unsigned int y = (unsigned char)(a + c);
unsigned int z = a * b;
...
```

$w = \frac{128}{7} \rightarrow 18.2 \rightarrow 128\%7 = r = 2 \rightarrow w = 2$

$x = 200 + 128 = 328$

$y =$ 328, but in unsigned char is 8 bits which ranges $[0, 255]$

$328 - 256 = 72 \rightarrow y = 72$

this means wrapping around.

$z = 200 \cdot 2 = 400$

$w = 2$ $x = 328$ $y = 72$ $z = 400$

Problem 2: Allocate space for the parameters of the BLAS sgemm function

The *Basic Linear Algebra Subroutines* (BLAS) library is a popular set of functions for implementing tensor operations. The "gemm" function is a BLAS subroutine that performs a "general matrix-matrix multiply". Specifically, it evaluates the expression:

$$\mathbf{C} = \alpha\mathbf{AB} + \beta\mathbf{C} \quad \text{where} \quad \mathbf{A} \in \mathbb{R}^{m \times k} \quad \text{and} \quad \mathbf{B} \in \mathbb{R}^{k \times n}$$

The data type for a BLAS function is specified by the first letter. For example, "dgemm" takes arrays representing 64-bit real values and "sgemm" takes arrays representing 32-bit real values. The C signature for the sgemm function looks like this:

```
void sgemm(int m, int n, int k, float alpha, float* A, float* B, float beta, float* C)
```

Write the code to allocate arrays for **A**, **B**, and **C** assuming all of the other values are specified.

```
#include <stdio.h>
#include <stdlib.h>
int m, n, K;
float alpha, beta;
float* A = (float*) malloc(m * k * sizeof(float));
float* B = (float*) malloc(k * n * sizeof(float));
float* C = (float*) malloc(m * n * sizeof(float));
```

Understand

$A = \mathbb{R}^{m \times k}$

$B = \mathbb{R}^{k \times n}$

$C = \mathbb{R}^{m \times n}$